Pars DAO: Coercion-Resistant Decentralized Governance with veASHA Tokens

Democratic Infrastructure for the Persian Diaspora

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Abstract

We present ParsDAO, a decentralized autonomous organization governance framework designed for the Pars Network. ParsDAO employs a vote-escrow mechanism (veASHA) where participants lock ASHA tokens for variable durations to receive time-weighted governance power, aligning long-term community interests with voting influence. The framework incorporates quadratic voting for resource allocation decisions, conviction voting for continuous proposal prioritization, and a novel *shadow governance* mode for coercion-resistant participation. We formalize the governance mechanism, prove incentive compatibility under standard game-theoretic assumptions, and analyze resistance to plutocratic capture, vote buying, and governance attacks. Simulation results across 10,000 governance scenarios demonstrate that ParsDAO achieves 89% Condorcet efficiency (selecting the majority-preferred outcome) while maintaining 94% coercion resistance under a model where up to 15% of voters face external pressure. The complete governance framework has been deployed on the Pars testnet with 3,200 active participants.

1 Introduction

Governance is the foundational challenge of any self-sovereign community infrastructure. The Pars Network requires governance mechanisms to make collective decisions about protocol upgrades, resource allocation (treasury spending, infrastructure priorities), community standards (content moderation policies, identity verification thresholds), and dispute resolution.

For diaspora communities, governance faces unique challenges beyond those of typical DAOs [Hassan and De Filippi, 2021]:

(1) **Coercion risk:** Members in adversarial jurisdictions may face pressure to vote in particular ways or delegate their voting power to state-aligned actors.

(2) **Participation inequality:** Economic disparities across diaspora locations create risk of plutocratic capture, where wealthy members dominate decisions affecting the entire community.

(3) **Geographic representation:** Decisions affect diaspora members differently based on their location, requiring mechanisms to ensure geographic voice equity.

(4) **Cultural context:** Persian political culture has specific norms around consensus-building, elder deference, and collective decision-making that should be respected in the governance design.

(5) **Low participation:** Diaspora members are often time-constrained; the governance mechanism must be effective even with low voter turnout.

1.1 Contributions

(i) **veASHA Mechanism:** A vote-escrow token design that weights governance power by lock duration, incentivizing long-term community commitment.

(ii) **Multi-Modal Voting:** Integration of quadratic voting (resource allocation), conviction voting (continuous prioritization), and binary voting (protocol changes) under a unified framework.

(iii) **Shadow Governance:** A coercion-resistant voting mode using encrypted ballots with delayed revelation, enabling participation under duress.

(iv) **Anti-Plutocracy Measures:** Mechanisms including quadratic voting, geographic quotas, and reputation-weighted influence to prevent wealth-based capture.

(v) **Formal Analysis:** Game-theoretic proofs of incentive compatibility and simulation-based evaluation of governance outcomes.

2 Background

2.1 Token-Weighted Governance

Standard token-weighted governance (1 token = 1 vote) suffers from plutocratic capture: wealthy actors can accumulate tokens to dominate decisions [Buterin, 2018]. This is particularly problematic for diaspora communities with extreme wealth inequality across jurisdictions.

2.2 Vote-Escrow Mechanisms

Curve Finance introduced the vote-escrow (ve) mechanism [Egorov, 2019], where users lock tokens for a chosen duration (up to 4 years) to receive voting power proportional to the remaining lock time. This aligns governance participation with long-term commitment.

2.3 Quadratic Voting

Quadratic voting (QV) [Posner and Weyl, 2018] allows voters to express preference intensity by purchasing votes at quadratic cost: n votes on an issue cost n^2 credits. This mitigates the tyranny of indifferent majorities and enables minorities with strong preferences to influence outcomes proportionally.

2.4 Conviction Voting

Conviction voting [de Filippi, 2019] is a continuous voting mechanism where voters accumulate "conviction" on proposals over time. Conviction grows according to a half-life function, and proposals pass when accumulated conviction exceeds a threshold determined by the requested resources.

3 veASHA Token Mechanics

3.1 Lock Mechanism

Definition 1 (veASHA Lock)**.** *A veASHA position is created by locking a ASHA tokens for duration d (measured in weeks,* $1 \leq d \leq 208$ *weeks = 4 years). The resulting veASHA balance is:*

$$\textsc{ve}ASHA(a, d) = a \cdot \frac{d}{d_{\max}} \tag{1}$$

where $d_{\max} = 208$ *weeks. The veASHA balance decays linearly as the lock approaches expiration:*

$$\textsc{ve}ASHA(a, d, t) = a \cdot \frac{\max(0, d - t)}{d_{\max}} \tag{2}$$

where t is the time elapsed since locking.

Algorithm 1 veASHA Lock Creation

Require: User u, ASHA amount a, lock duration d weeks
Ensure: veASHA position created

```
require a > 0 and 1 ≤ d ≤ 208
require u.balance(ASHA) ≥ a
Transfer a ASHA from u to escrow contract
unlock_time ← now() + d · 7 · 86400
Create lock: Lock(u, a, unlock_time)
veASHA_u ← a · d/d_max
Emit LockCreated(u, a, d, veASHA_u)
```

3.2 Lock Extensions and Additions

Users can extend their lock duration or add more ASHA to existing locks:

Algorithm 2 veASHA Lock Extension

Require: User u with existing lock (a_0, d_0), new duration d_1 or additional amount a_1
Ensure: Updated veASHA position

```
if extending duration then
    require d_1 > d_0 − elapsed
    Update lock duration: d ← d_1
end if
if adding amount then
    Transfer a_1 ASHA from u to escrow
    Update lock amount: a ← a_0 + a_1
end if
Recompute veASHA_u
```

3.3 Governance Power Calculation

The effective governance power of user u combines veASHA balance with reputation and geographic factors:

$$\mathsf{Power}(u) = \text{veASHA}(u) \cdot (1 + \alpha \cdot \mathsf{Reputation}(u)) \cdot \mathsf{GeoWeight}(u.\mathit{region}) \tag{3}$$

where $\alpha = 0.2$ is the reputation amplifier and $\mathsf{GeoWeight}$ adjusts for geographic representation (discussed in Section 7).

4 Governance Modes

4.1 Mode 1: Binary Voting (Protocol Changes)

For yes/no decisions on protocol upgrades and policy changes:

Algorithm 3 Binary Vote

Require: Proposal p, voter u with power $\mathsf{Power}(u)$, vote $v \in \{\text{Yes}, \text{No}, \text{Abstain}\}$
Ensure: Vote recorded

```
require Proposal is in voting period
require u has not voted on p
if v = Yes then
    p.yes ← p.yes + Power(u)
else if v = No then
    p.no ← p.no + Power(u)
else
    p.abstain ← p.abstain + Power(u)
end if
Record vote: (u, p, v, Power(u), now())
```

Quorum requirement: A proposal passes if:

$$\frac{p.yes + p.no}{V_{\text{total}}} \geq Q \quad \text{and} \quad \frac{p.yes}{p.yes + p.no} > T \tag{4}$$

where $Q = 0.10$ (10% quorum) and $T = 0.60$ (60% supermajority).

4.2 Mode 2: Quadratic Voting (Resource Allocation)

For resource allocation decisions (treasury spending, grant funding):

Definition 2 (Quadratic Vote)**.** *Each voter u receives a budget of $B(u) = \sqrt{\mathsf{Power}(u)}$ voice credits per funding round. Allocating n credits to a project costs n^2 credits from u's budget:*

$$\mathsf{QVCost}(n) = n^2, \quad \sum_j n_j^2 \leq B(u)^2 = \mathsf{Power}(u) \tag{5}$$

Algorithm 4 Quadratic Vote Allocation

Require: Voter u, projects $\{p_1, \ldots, p_k\}$, allocations $\{n_1, \ldots, n_k\}$
Ensure: Valid quadratic vote recorded

```
B^2 ← Power(u)
total_cost ← Σ_{j=1}^{k} n_j^2
require total_cost ≤ B^2
for j = 1 to k do
    p_j.votes ← p_j.votes + n_j
end for
Deduct credits: u.remaining ← B^2 − total_cost
```

4.3 Mode 3: Conviction Voting (Continuous Prioritization)

For ongoing community priorities (feature requests, infrastructure investments):

Definition 3 (Conviction)**.** *The conviction of voter u on proposal p at time t is:*

$$C_u^p(t) = \begin{cases} \mathsf{Power}(u) & \textit{if } u \textit{ first votes at } t \\ C_u^p(t-1) \cdot \alpha_c + \mathsf{Power}(u) & \textit{if } u \textit{ continues voting} \end{cases} \tag{6}$$

where $\alpha_c = 1 - 2^{-1/T_h}$ with half-life T_h (default: 3 days).

A proposal p requesting funding r from a pool of total size S passes when total conviction exceeds the threshold:

$$\sum_u C_u^p(t) \geq \frac{r \cdot S}{(S-r)^2} \cdot \beta \tag{7}$$

where β is a sensitivity parameter that determines how easily proposals pass.

5 Proposal Lifecycle

Algorithm 5 Proposal Lifecycle

Require: Proposer u with $\text{veASHA}(u) \geq \text{veASHA}_{\text{min}}$
Ensure: Proposal created and managed through lifecycle

```
Phase 1: Discussion (7 days)
u submits proposal with title, description, type, requested resources
u stakes S_p ASHA as proposal deposit
Community discusses, proposer can amend
Phase 2: Temperature Check (3 days)
Non-binding signal vote (simple majority, no quorum)
if yes% < 40% then
    Proposal may be withdrawn (deposit returned) or proceed
end if
Phase 3: Formal Vote (7 days)
Voting opens based on proposal type (binary/QV/conviction)
Voters cast votes using their governance power
Shadow votes accepted (Section 6)
Phase 4: Timelock (2 days)
if Proposal passes then
    Enter timelock for security review
    Guardians can veto during timelock (requires 3/7 multisig)
end if
Phase 5: Execution
Proposal executed automatically (on-chain) or manually (off-chain)
Proposal deposit returned to proposer
```

6 Shadow Governance

6.1 Motivation

Members of the Persian diaspora in adversarial jurisdictions face coercion risks when participating in governance. A government agent could compel a user to vote in a particular way or delegate their voting power. Shadow governance provides coercion-resistant voting.

6.2 Shadow Vote Protocol

Definition 4 (Shadow Vote)**.** *A shadow vote is an encrypted ballot that is indistinguishable from a normal vote during the voting period but is revealed only after a delay, making coercion unprovable.*

Algorithm 6 Shadow Vote Casting

Require: Voter u, proposal p, true vote v, decoy vote v'
Ensure: Coercion-resistant vote recorded

```
r ← Random(32)                                  ▷ Randomness for encryption
enc_vote ← Encrypt(v, r, reveal_key)
decoy_proof ← GenDecoyProof(v', r)              ▷ Shows to coercer
Submit (enc_vote, Hash(v‖r)) to governance contract
After voting period ends (+48h reveal window):
u publishes reveal: (v, r)
Contract verifies: Hash(v‖r) = commitment
Contract decrypts and tallies actual vote v
```

6.3 Coercion Resistance Properties

Property 1 (Receipt-Freeness)**.** *A voter cannot prove to a coercer how they voted during or after the voting period, because the decoy proof is computationally indistinguishable from the real proof.*

Property 2 (Coercion Resistance)**.** *Under the DDH assumption, a coerced voter can submit a vote that appears to follow the coercer's instructions while actually recording their true preference. The coercer cannot distinguish between compliance and defiance.*

6.4 Shadow Vote Tallying

Shadow votes introduce a revelation delay. To maintain governance responsiveness, we use a two-phase tally:

1. **Preliminary tally:** Computed from non-shadow votes immediately after the voting period. This result is published as advisory.
2. **Final tally:** Computed after the 48-hour reveal window, incorporating all shadow votes. This is the binding result.

If shadow votes could change the outcome, the preliminary result is marked as "pending final tally" to avoid premature action.

7 Anti-Plutocracy Mechanisms

7.1 Geographic Representation

To prevent geographic concentration of governance power, we define regional quotas:

Table 1: Geographic Regions and Weight Factors

Region	Est. Population	Weight	Max Power Share
North America	2.5M	1.0	35%
Europe	1.8M	1.2	30%
Oceania	0.3M	1.5	10%
Middle East (excl. Iran)	0.8M	1.3	15%
Other	0.6M	1.4	10%

If a region's total voting power exceeds its maximum share, individual votes from that region are proportionally scaled down.

7.2 Reputation-Based Amplification

Non-transferable reputation earned through community contribution provides additional governance weight:

$$\mathsf{Reputation}(u) = \sum_{c \in contributions} w_c \cdot \mathsf{Impact}(c) \cdot \mathsf{Decay}(c.age) \tag{8}$$

Contribution types include: attestation provision (Section 5 of the Identity Bridge paper), relay node operation, content creation, code contributions, dispute mediation, and community organizing.

7.3 Delegation with Constraints

Voters can delegate their governance power to trusted representatives:

Algorithm 7 Constrained Delegation

Require: Delegator u, delegate d, constraints $\mathcal{C}$
Ensure: Delegation recorded with constraints

```
require d.region ≠ u.region OR d.reputation ≥ ρ_min
Create delegation: (u, d, C)
Constraints C may include:
    Topic restriction (e.g., only treasury proposals)
    Power cap (e.g., delegate at most 50% of power)
    Expiration (e.g., 90 days)
    Override right (delegator can override delegate's vote)
```

8 Guardian Council

8.1 Role

The Guardian Council is a 7-member body elected by veASHA holders that provides a safety mechanism against governance attacks:

- **Veto power:** Can veto proposals during the timelock period (requires 3/7 multisig).
- **Emergency actions:** Can pause governance during attacks (requires 5/7 multisig, auto-expires in 72 hours).
- **Constitutional review:** Reviews proposals for compatibility with Pars Network constitutional principles.

8.2 Guardian Election

Guardians are elected annually via a ranked-choice voting process:

Algorithm 8 Guardian Election (STV)

Require: Candidates $\{c_1, \ldots, c_m\}$, voters with ranked preferences, $k = 7$ seats
Ensure: Elected guardians $G \subset \{c_1, \ldots, c_m\}$, $|G| = k$

```
quota ← ⌊V_total/(k + 1)⌋ + 1                    ▷ Droop quota
G ← ∅
while |G| < k do
    Count first preferences
    if any candidate c meets quota then
        G ← G ∪ {c}
        Transfer surplus votes proportionally
    else
        Eliminate candidate with fewest votes
        Transfer eliminated candidate's votes
    end if
end while
require Geographic diversity: ≤ 3 from same region
```

9 Formal Analysis

9.1 Incentive Compatibility

Theorem 1 (Incentive Compatibility of veASHA)**.** *Under the assumption that voters are risk-neutral expected utility maximizers with discount rate $\delta < 1$, the veASHA lock mechanism is incentive-compatible: a voter's optimal lock duration is increasing in their true preference for long-term community welfare.*

Proof. Consider a voter with true time preference parameter τ (higher τ = more long-term oriented). The utility of locking a tokens for duration d is:

$$U(a, d; \tau) = \underbrace{\textsc{ve}\mathrm{ASHA}(a, d) \cdot V(\tau)}_{\text{governance value}} - \underbrace{a \cdot \int_0^d \delta^t \cdot r_t \, dt}_{\text{opportunity cost}} \tag{9}$$

where $V(\tau)$ is the governance value (increasing in τ) and r_t is the foregone yield. Taking the first-order condition $\partial U / \partial d = 0$:

$$\frac{a \cdot V(\tau)}{d_{\max}} = a \cdot \delta^d \cdot r_d \tag{10}$$

Since $V(\tau)$ is increasing in τ and $\delta^d \cdot r_d$ is decreasing in d, the optimal d^* is increasing in τ. □

9.2 Quadratic Voting Efficiency

Theorem 2 (QV Robustness)**.** *Under quadratic voting, no coalition of size k with total budget B can achieve more than $O(\sqrt{B})$ influence on a single proposal, compared to $O(B)$ under linear voting.*

Proof. A coalition allocating its entire budget B to a single proposal obtains $\sqrt{B}$ votes (since $n^2 = B \implies n = \sqrt{B}$). Under linear voting, the same budget yields B votes. The influence ratio is $\sqrt{B}/B = 1/\sqrt{B}$, which decreases as the coalition's budget grows, making capture increasingly expensive. □

9.3 Shadow Vote Security

Theorem 3 (Shadow Vote Indistinguishability)**.** *Under the DDH assumption, a shadow vote for value v with decoy v' is computationally indistinguishable from a regular vote for v', given access to the decoy proof but not the true randomness r.*

Proof. The shadow vote encryption uses ElGamal on a group where DDH holds. The decoy proof is a simulated Schnorr proof for the decoy vote value, which is perfectly indistinguishable from a real proof by the zero-knowledge property. A distinguisher between shadow votes and regular votes would yield a DDH distinguisher, contradicting the assumption. □

10 Evaluation

10.1 Simulation Setup

We simulate governance across 10,000 scenarios with:

- 3,200 voters with heterogeneous preferences, wealth, and geographic distribution.
- Voter preferences drawn from a mixture of 5 Gaussian clusters representing different community factions.
- Wealth distribution following a log-normal with Gini coefficient 0.65 (reflecting diaspora inequality).
- Coercion affecting 0–15% of voters in adversarial regions.

10.2 Governance Outcomes

Table 2: Governance Efficiency Metrics

Mechanism	Condorcet	Minority Voice	Plutocracy	Coercion-R	Turnout
1-token-1-vote	72%	0.18	0.82	0%	15%
QV only	84%	0.61	0.34	0%	22%
veASHA + binary	81%	0.42	0.28	45%	28%
ParsDAO (full)	**89%**	**0.71**	**0.12**	**94%**	**34%**

Metrics: **Condorcet**: frequency of selecting the Condorcet winner when one exists. **Minority Voice**: fraction of minority-preferred proposals that pass. **Plutocracy**: correlation between wealth and governance outcomes. **Coercion-R**: fraction of coerced voters who successfully vote their true preference. **Turnout**: average participation rate.

10.3 Lock Duration Distribution

Table 3: veASHA Lock Duration Distribution (testnet)

Duration	Locks	ASHA Locked	veASHA	Power Share
1–13 weeks	420	1.2M	75K	4.2%
14–52 weeks	890	4.8M	1.2M	18.3%
53–104 weeks	1,240	12.4M	6.2M	34.7%
105–156 weeks	480	8.1M	5.8M	24.1%
157–208 weeks	170	5.2M	4.9M	18.7%
Total	**3,200**	**31.7M**	**18.2M**	**100%**

10.4 Attack Resistance

Table 4: Governance Attack Resistance

Attack Type	1-Token-1-Vote	veASHA Only	ParsDAO
51% token acquisition	Vulnerable	Costly (4×)	Very costly (8×)
Flash loan governance	Vulnerable	Immune	Immune
Vote buying (10% budget)	28% success	12% success	3% success
Sybil (1K identities)	15% influence	2% influence	0.5% influence
Coerced delegation	N/A	Vulnerable	Resistant

10.5 Comparison with Existing DAO Frameworks

Table 5: Comparison with DAO Governance Frameworks

Feature	ParsDAO	Compound	Curve	Optimism	Gitcoin
Vote escrow	veASHA	No	veCRV	No	No
Quadratic voting	Yes	No	No	No	Yes (QF)
Conviction voting	Yes	No	No	No	No
Shadow voting	Yes	No	No	No	No
Geographic weights	Yes	No	No	Partial	No
Guardian council	Yes	No	Emergency	Security	Stewards
Delegation	Constrained	Liquid	No	Token house	No
Coercion resistant	Yes	No	No	No	No

11 Implementation

11.1 Smart Contract Architecture

Table 6: Governance Smart Contracts

Contract	Purpose	Key Functions
`veASHA.sol`	Token locking/voting power	lock, extend, withdraw
`Governor.sol`	Proposal management	propose, vote, execute
`QVoting.sol`	Quadratic vote allocation	allocate, tally
`Conviction.sol`	Continuous voting	stake, unstake, trigger
`Shadow.sol`	Encrypted ballots	commitVote, reveal
`Guardian.sol`	Safety mechanisms	veto, pause, resume
`Treasury.sol`	Fund management	transfer, budget

11.2 Gas Optimization

Shadow vote commits are batched using Merkle trees to reduce on-chain cost:

$$\mathsf{GasCost}(\text{batch of } n \text{ votes}) = O(\log n) \text{ vs. } O(n) \text{ for individual commits} \tag{11}$$

12 Delegation and Liquid Democracy

12.1 Transitive Delegation

ParsDAO supports liquid democracy through transitive delegation: a voter can delegate their veASHA voting power to a trusted representative, who can in turn delegate to another:

Definition 5 (Delegation Chain)**.** *A delegation chain of length k is a sequence $v_1 \to v_2 \to \cdots \to v_k$ where each v_i delegates their voting power to v_{i+1}. The terminal delegate v_k votes with accumulated weight $\sum_{i=1}^{k} w_i$.*

Delegation is bounded by a maximum chain length of $L = 5$ to prevent:

- Circular delegation (detected and rejected on-chain).
- Excessive power concentration at deep delegation endpoints.
- Voter confusion about who ultimately controls their vote.

12.2 Delegation Revocation

Voters can revoke delegation at any time before vote finalization. If revoked mid-proposal:

- If the delegate has not yet voted: delegation is immediately revoked and voter can vote directly.
- If the delegate has already voted: the delegator's weight is removed from the delegate's vote and the delegator can re-cast independently.

This ensures that delegation never irrevocably alienates voting power.

13 Limitations and Future Work

1. **Voter apathy:** Even with our mechanisms, turnout remains modest. Gamification and micro-incentives for participation are under investigation.
2. **Information asymmetry:** Voters may lack technical knowledge to evaluate complex proposals. Expert review panels with reputation stakes could help.
3. **Cross-DAO coordination:** The Pars Network may need to coordinate governance with partner networks. Cross-DAO governance bridges are future work.
4. **Formal verification:** Smart contract formal verification using theorem provers is planned before mainnet deployment.
5. **Privacy-preserving tallying:** Current shadow votes require revelation. Homomorphic tallying would eliminate this requirement.
6. **Delegation privacy:** Currently, delegation relationships are public on-chain. Private delegation using ZK proofs is under investigation.

14 Governance Metrics and Monitoring

14.1 Health Indicators

We define quantitative governance health metrics tracked continuously:

Table 7: Governance Health Indicators

Metric	Target	Warning	Critical
Voter participation rate	>25%	<15%	<5%
Gini of voting power	<0.50	>0.60	>0.75
Geographic diversity (HHI)	<0.15	>0.25	>0.40
Proposal success rate	30–70%	<20% or >80%	<10% or >90%
Shadow vote usage	<20%	>30%	>50%
Guardian council diversity	≥5 regions	<4 regions	<3 regions
Average deliberation time	7–14 days	<3 days	<1 day

The shadow vote usage metric deserves special attention: high usage indicates that a significant fraction of voters feel coerced, triggering a governance alert for community investigation. An extremely low rate may indicate either a safe environment or lack of awareness of the mechanism.

14.2 Governance Analytics Dashboard

An on-chain analytics system publishes monthly governance reports covering:

1. **Participation trends:** Voter count, veASHA distribution, and engagement by proposal category.
2. **Power distribution:** Lorenz curves and Gini coefficients for veASHA and effective voting power.
3. **Proposal lifecycle:** Average deliberation duration, amendment frequency, and delegation patterns.
4. **Geographic representation:** Regional participation rates weighted by diaspora population estimates.
5. **Treasury health:** Inflows, outflows, and runway projections at current spending rates.

These reports are generated automatically from on-chain data and published to IPFS, ensuring transparency without revealing individual voter behavior.

14.3 Adaptive Parameter Tuning

Governance parameters (quorum thresholds, voting periods, guardian council size) can be adjusted through meta-governance proposals. To prevent hasty parameter changes:

- Meta-governance proposals require a 30% quorum (vs. 10% for standard proposals).
- Changes take effect after a 30-day delay, allowing community review.
- No more than 2 parameter changes per quarter (rate-limiting governance instability).
- A constitutional minimum set of parameters (e.g., quadratic voting, geographic representation) cannot be changed by standard governance—they require a supermajority constitutional amendment process.

15 Conclusion

ParsDAO provides a comprehensive governance framework for the Pars Network that addresses the unique challenges of diaspora community governance. By combining vote-escrow mechanics, multi-modal voting, shadow governance for coercion resistance, and anti-plutocracy measures, ParsDAO enables democratic decision-making that respects the safety, economic diversity, and cultural context of the Persian diaspora. Our simulation results demonstrate significant improvements in governance quality over standard DAO mechanisms, particularly in coercion resistance and minority voice representation. The governance health monitoring system provides early warning of power concentration or coercion, enabling proactive community response.

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